

The Stored-Program Computer: From Theory to Transistors

AutoGenBook benchmark corpus

1 A theory of computation

In 1936 the British mathematician Alan Turing published a paper on computable numbers in which he described an abstract machine that reads and writes symbols on an unbounded tape according to a finite table of rules. Turing showed that a single universal machine could imitate any other such machine if the description of that machine was written on its tape. The universal machine anticipated the idea that a program is itself a kind of data. In the same year the American logician Alonzo Church reached equivalent conclusions with a different formalism, and the claim that these models capture everything that can be computed mechanically became known as the Church–Turing thesis.

2 Electronic machines of the war years

The first electronic digital devices used vacuum tubes as switches, because tubes could change state thousands of times faster than the relays of electromechanical calculators. Between 1939 and 1942 John Atanasoff and Clifford Berry built at Iowa State College a machine for solving systems of linear equations. It computed in binary and stored intermediate values on rotating drums of capacitors, but it was not programmable.

In Britain, the engineer Tommy Flowers designed Colossus to help break the Lorenz cipher used by the German high command. The first Colossus was operational at Bletchley Park in early 1944. It was programmed by switches and plugs, and ten machines were in use by the end of the war. Their existence remained secret until the 1970s.

3 ENIAC

ENIAC, the Electronic Numerical Integrator and Computer, was built at the University of Pennsylvania by John Mauchly and J. Presper Eckert and was completed in 1945. It was designed to calculate artillery firing tables for the United States Army. The machine contained about 17,468 vacuum tubes, weighed around 27 tonnes and occupied a large room.

ENIAC was fast, performing about five thousand additions per second, but it had no stored program. To change the problem it solved, operators had to reconnect cables on plugboards and set thousands of switches, a process that could take days. The women who programmed it, among them Kathleen McNulty, Jean Jennings and Frances Elizabeth Holberton, worked out the wiring from the mathematical description of each problem.

4 The EDVAC report and the stored-program concept

Even before ENIAC was finished, its designers planned a successor that would avoid the rewiring problem. In June 1945 the mathematician John von Neumann, who had joined the discussions as a consultant, wrote the First Draft of a Report on the EDVAC. The report described a machine whose instructions were held in the same memory as the data, encoded

as numbers. A program could then be loaded as quickly as data, and the machine could even modify its own instructions.

The organisation described in the report consisted of a central arithmetic unit, a control unit, a memory, and input and output devices. It became known as the von Neumann architecture, although the ideas were developed jointly with Eckert and Mauchly, who disputed the attribution. The separation between a processor that fetches instructions and a memory that holds both instructions and data is still the basis of most computers.

5 The first stored programs

The first machine to run a program stored in its electronic memory was the Manchester Small-Scale Experimental Machine, nicknamed the Baby, built by Frederic Williams, Tom Kilburn and Geoff Tootill at the University of Manchester. On 21 June 1948 it ran a program that found the highest proper factor of a number. Its memory was a cathode-ray tube that stored bits as charged spots on the screen, a device known as the Williams tube.

The first stored-program computer to provide a regular computing service was EDSAC, built at the University of Cambridge under the direction of Maurice Wilkes. It ran its first program in May 1949. The EDSAC team introduced a library of subroutines and a simple assembler called initial orders, which translated mnemonic instructions into machine code when a program was loaded.

Machine	Year	Switching technology	Programming method
Colossus	1944	Vacuum tubes	Switches and plugs
ENIAC	1945	Vacuum tubes	Plugboards and switches
Manchester Baby	1948	Vacuum tubes	Stored program
EDSAC	1949	Vacuum tubes	Stored program with subroutines

Table 1: Early electronic computers

6 From vacuum tubes to transistors

Vacuum tubes were bulky, consumed a lot of power and failed often. In December 1947 John Bardeen and Walter Brattain, working in the group of William Shockley at Bell Laboratories, demonstrated the point-contact transistor, a solid-state amplifier and switch made of germanium. The three received the Nobel Prize in Physics in 1956. By the late 1950s transistors had replaced tubes in new computers, which became smaller, cheaper and far more reliable.

The next step was to place many transistors on a single piece of semiconductor. In 1958 Jack Kilby at Texas Instruments built the first integrated circuit from germanium, and in 1959 Robert Noyce at Fairchild Semiconductor developed a silicon integrated circuit using the planar process, which made mass production practical. In 1965 Gordon Moore observed that the number of components on an integrated circuit was doubling at a regular rate, an observation that became known as Moore's law and guided the semiconductor industry for decades.

7 Why the stored program mattered

The stored-program concept turned the computer from a configurable calculator into a universal machine in Turing's sense. Because programs were data, they could be written, copied, loaded and translated by other programs. Assemblers, compilers and operating systems all depend on this property. The history of software, from the subroutine library of EDSAC to modern programming languages, begins with the decision to keep instructions in memory.